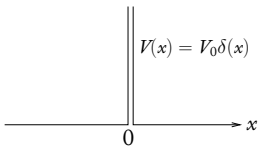
Problem Set 4

Due Saturday, October 3, 11:59 PM.

You may use Mathematica or other computational tools in all problem sets.

1. Consider a particle of mass of mass m and $E > 0$ is incident from the left toward the origin in the potential shown in the figure below with $V(x) = V_0\delta(x)$, where $V_0 > 0$.

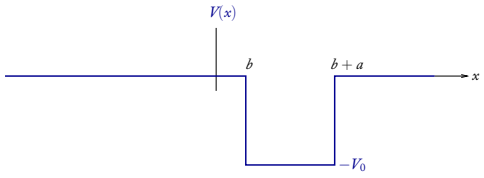


   (a) Determine the ratio of the amplitudes of the wave transmitted to the right to that of the incident and the ratio of the amplitudes of the wave reflected back to the left to that of the incident. (5 points)

   (b) Make schematic plots of the transmission and the reflection coefficients and their sum as a function of energy E. Do the plots make sense? (5 points)

2. (a) Single and double potential wells. Answer the following questions qualitatively without doing calculations.

      i. Make a schematic plot of the ground state wavefunction for a bound particle in the single potential well of width a and depth V_0 below. Denote the wavefunction by $\psi(x)$ and the energy by E. (5 points)

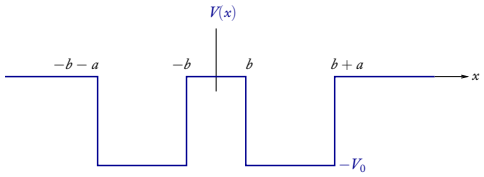


      ii. Make a schematic plot of $E + V_0$ as a function of V_0, from $V_0 = 0$ to $V_0 = \infty$, with a fixed. Mark the values at $V_0 = 0$ to $V_0 = \infty$. (5 points)

      iii. Make a schematic plot of $E + V_0$ as a function of a, from $a \approx 0$ to $a = \infty$, with V_0 fixed. (5 points)

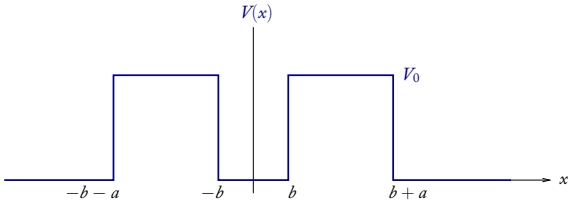
      iv. Make a schematic plot of the ground state and the first exited state wavefunctions, denote them by $\psi_s(x)$ and $\psi_a(x)$, for a bound particle in the double potential well below.

Assume that the wells are deep enough that each well can be separately viewed as close to (but not exactly) an infinite square well for the lower lying states. Point out how $\psi_s(x)$ and $\psi_a(x)$ are qualitatively related to the $\psi(x)$ of a similarly deep well in part (2(a)i). Would the corresponding energies E_s and E_a be equal, greater, or smaller than E of the single well. (5 points) [Double well potentials are used for various physical systems such as H_2 molecule.]



v. Make a schematic plot of both $E_s + V_0$ and $E_a + V_0$ as a function of b, from $b = 0$ to $b = \infty$, and point out how these are related to E. (5 points)

(b) Now the double potential well is inverted upside down and converted to the double potential barrier shown in the figure below.



i. Estimate the lifetime τ of a bound particle of mass m and energy E between the barriers in terms of given parameters. (5 points) [Assume that E and the depth and the width of the barriers are such that the tunneling formula, which gives the probability for tunneling $T = e^{-2\int_b^{b+a}\sqrt{\frac{2m(V_0-E)}{\hbar^2}}dx}$ for the barrier on the right, applies.]

ii. Make a schematic plot of τ as a function of V_0, from $V_0 = 0$ to $V_0 = \infty$, with a fixed and $E \approx 0$. Make a second schematic plot of τ as a function of a, from $a = 0$ to $a = \infty$, with V_0 fixed. (5 points)

3. (a) Starting with $\psi(\xi) = H(\xi)e^{-\frac{1}{2}\xi^2}$, $H(\xi) = \sum_{n=0}^{\infty} a_n \xi^n$, and the recursion relation $a_{n+2} = \frac{2n+1-K}{(n+2)(n+1)} a_n$ with $\xi \equiv \sqrt{\frac{m\omega}{\hbar}}\, x$ and $K \equiv \frac{2E}{\hbar\omega}$, determine the energies and the corresponding wavefunctions for the second ($n = 2$) and the third ($n = 3$) excited states of a particle of mass m in a simple harmonic oscillator with potential $V(x) = \frac{1}{2}m\omega^2 x^2$. (5 points)

(b) Starting with the Rodrigues formula for $H(\xi)$, $H_n(\xi) = (-1)^n e^{\xi^2}\left(\frac{d}{d\xi}\right)^n e^{-\xi^2}$, determine the wavefunctions for the second ($n = 2$) and the third ($n = 3$) excited states, check that the results are the same as in part (3a) (up to irrelevant overall constant factors). (5 points)

(c) Determine the uncertainties for position Δx and momentum Δp for a particle in the ground state of the simple harmonic oscillator, and check whether the uncertainty principle holds. (5 points)

(d) The particle is in state $\psi(x,0) = \frac{1}{\sqrt{2}}(\psi_0(x) + \psi_1(x))$ of the simple harmonic oscillator at time $t = 0$.

i. Write down the wavefunction $\psi(x,t)$ at time t. Set $\hbar = m = \omega = 1$ and use Mathematica to plot $|\psi(x,t)|^2$ at $t = 0$, $t = \frac{\pi}{2}$, $t = \pi$, $t = \frac{3\pi}{2}$, and $t = 2\pi$. (5 points)

ii. Determine $\langle \hat{x} \rangle$ at time t, and find the amplitude and frequency of oscillation of $\langle \hat{x} \rangle$. (5 points)

iii. Determine $\langle \hat{p} \rangle$ at time t. (5 points)

4. How long did it take you to complete this problem set?